OpenADMET PXR Agonism Challenge

Model Performance Overview Report

Evaluated on 253 Revealed Test Compounds

2026-05-29

Revealed test compounds	253 (true pEC50 from pxr_test_unblinded.csv)
Baseline MAE (predict mean)	0.7987 (RAE denominator)
Individual models evaluated	157
Models with $R^2 > 0.3$	90
Models excluded ($R^2 \leq 0.3$)	58
Ensemble models ($R^2 > 0.3$)	9
Best individual model	tabicl_chemeleon_hts_mtl_pruned50 → RAE=0.5825 R^2=0.565 ρ=0.783
Best ensemble	ensemble_enet_34model_gated_t30 → RAE=0.5182 R^2=0.666 ρ=0.839

Evaluation protocol: All metrics are computed against the 253 true pEC50 values released post-submission. The test set was fully blinded at prediction time. RAE (Relative Absolute Error) = MAE / mean-absolute-deviation from the mean; RAE < 1.0 means the model beats naive mean prediction. Models with $R^2 \leq 0.3$ are excluded — they show insufficient correlation with the true labels to be informative.

1. Individual Model Rankings

Top 25 individual models by RAE, out of 90 qualifying models ($R^2 > 0.3$). All model predictions are test-set predictions on 513 test compounds, evaluated only on the 253 revealed subset.

Model	Family	MAE	RAE	R^2	Spearman
tabicl_chemeleon_hts_mtl_pruned50	TabICL	0.4653	0.5825	0.565	0.783
tabpfn_v3_chemeleon_hts_mtl	TabPFN	0.4677	0.5856	0.566	0.789
tabicl_chemeleon_hts_v2_ecfp4rd	TabICL	0.4695	0.5879	0.587	0.792
tabicl_chemeleon_hts_mtl	TabICL	0.4701	0.5886	0.544	0.795
tabicl_chemeleon_hts_v2	TabICL	0.4725	0.5916	0.574	0.779
tabicl_chemeleon_hts_cont	TabICL	0.4746	0.5943	0.534	0.786
graphgps_v3	GraphGPS	0.4750	0.5948	0.522	0.808
tabicl_chemeleon_hts_cont_pruned50	TabICL	0.4789	0.5996	0.542	0.780
tabpfn_v3_chemeleon_hts_cont	TabPFN	0.4792	0.6000	0.541	0.777
tabicl_chemeleon_hts	TabICL	0.4806	0.6018	0.529	0.790
chemprop_hts_official_mae_wt	ChemProp	0.4819	0.6034	0.509	0.775
tabpfn_v3_chemeleon_hts	TabPFN	0.4833	0.6052	0.527	0.793
chemprop_hpo	ChemProp	0.4840	0.6060	0.503	0.765
tabicl_chemeleon_hts_pruned50	TabICL	0.4851	0.6074	0.531	0.788
tabpfn_v3_mole	TabPFN	0.4857	0.6082	0.439	0.758
tabicl_vanilla_hts	TabICL	0.4887	0.6118	0.502	0.777
xgb_moe_rdkit2d	XGBoost	0.4996	0.6256	0.486	0.772
xgb_moe_rdkit_ecfp4	XGBoost	0.5009	0.6272	0.466	0.774
xgb_moe_ecfp4	XGBoost	0.5039	0.6309	0.454	0.772
tabicl_chemeleon_ecfp4_warm	TabICL	0.5046	0.6318	0.521	0.787
lgb_moe_rdkit_ecfp4	LightGBM	0.5048	0.6320	0.455	0.763
mole_e50	MolE	0.5063	0.6339	0.399	0.748
tabicl_admet_ai_pruned50	TabICL	0.5065	0.6342	0.441	0.774
tabicl_admet_ai	TabICL	0.5072	0.6351	0.448	0.778
lgb_moe_rdkit2d	LightGBM	0.5099	0.6385	0.471	0.758

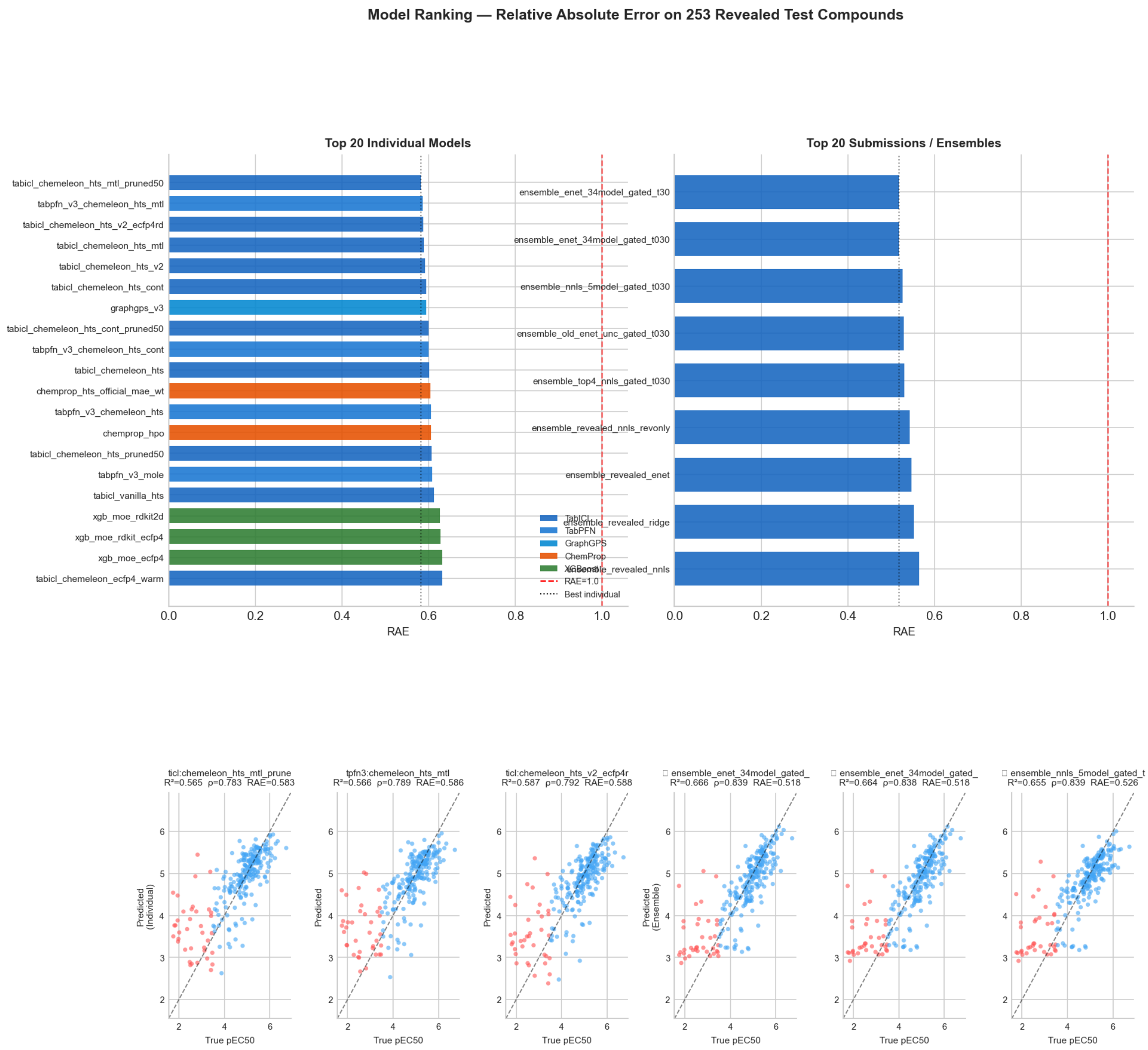

Figure 1. Left: top 20 individual models by RAE, colored by model family. Right: all ensemble/submission files. Red dashed = mean baseline (RAE=1.0). Dotted = best individual.

2. Per-Family Performance

Families are ranked by their best model's RAE. 'N' = number of qualifying variants (R^2>0.3) within the family. The range (min–max) shows performance spread across all configurations tested.

Family	N	Best RAE	Mean RAE	Best R^2	Best ρ
TabICL	22	0.5825	0.6515	0.587	0.795
TabPFN	17	0.5856	0.6572	0.566	0.793
GraphGPS	3	0.5948	0.6518	0.522	0.808
ChemProp	8	0.6034	0.6661	0.509	0.775
XGBoost	6	0.6256	0.6506	0.486	0.774
LightGBM	23	0.6320	0.6736	0.471	0.763
MolE	3	0.6339	0.6550	0.399	0.761
CheMeleon	1	0.6406	0.6406	0.488	0.772
ADMET-AI	1	0.6512	0.6512	0.431	0.756
Uni-Mol	1	0.6830	0.6830	0.317	0.755
CatBoost	1	0.6924	0.6924	0.374	0.729
CLAMP	1	0.7199	0.7199	0.346	0.703
GP	2	0.7297	0.7352	0.416	0.708
DANN	1	0.7395	0.7395	0.399	0.652

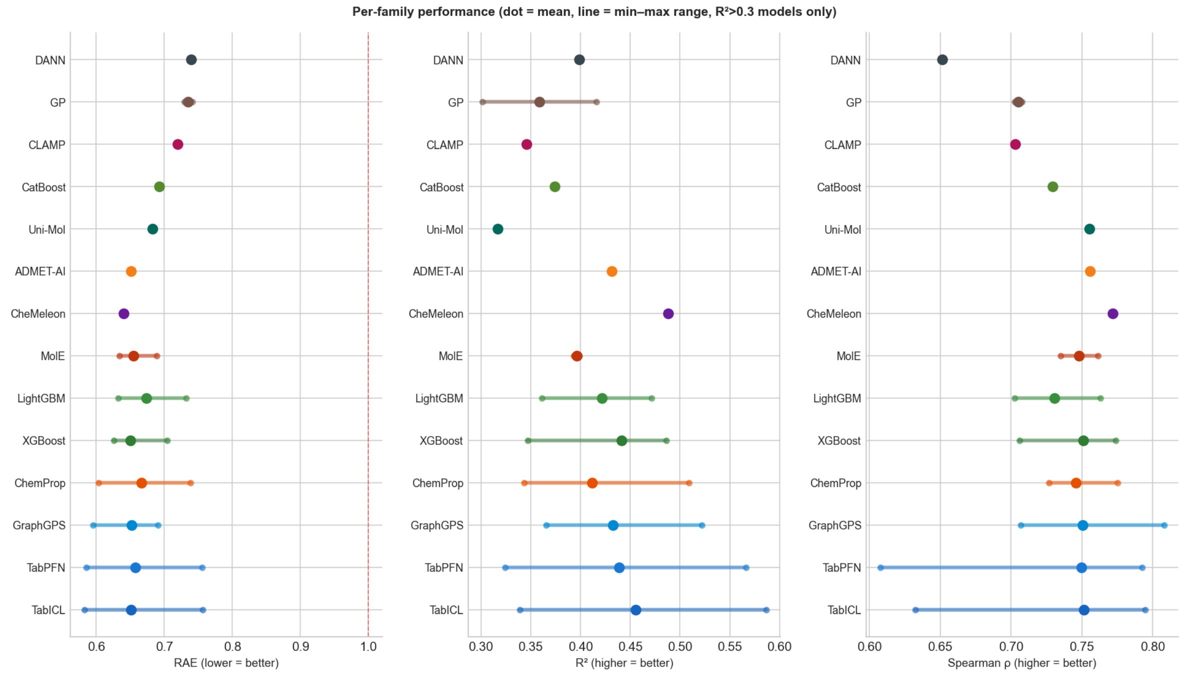

Figure 2. Per-family RAE, R^2, and Spearman range. Dot = family mean; line spans min–max. Red dashed = RAE=1.0.

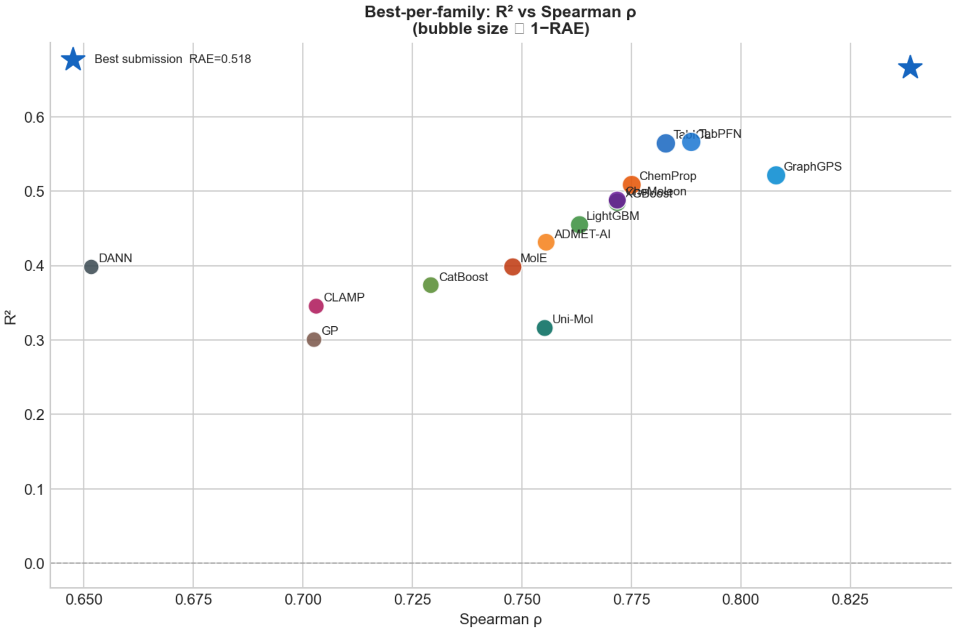

Figure 3. Best-per-family R^2 vs Spearman ρ. Bubble size proportional to (1–RAE). Star = best ensemble submission.

3. Top Individual Models — Predicted vs True pEC50

Scatter plots for the six best-performing individual models. Blue = active compounds (pEC50 ≥ 3.5); red = inactive. Identity line dashed. The best models come from the TabICL and TabPFN families using CheMeleon embeddings pre-trained on HTS data.

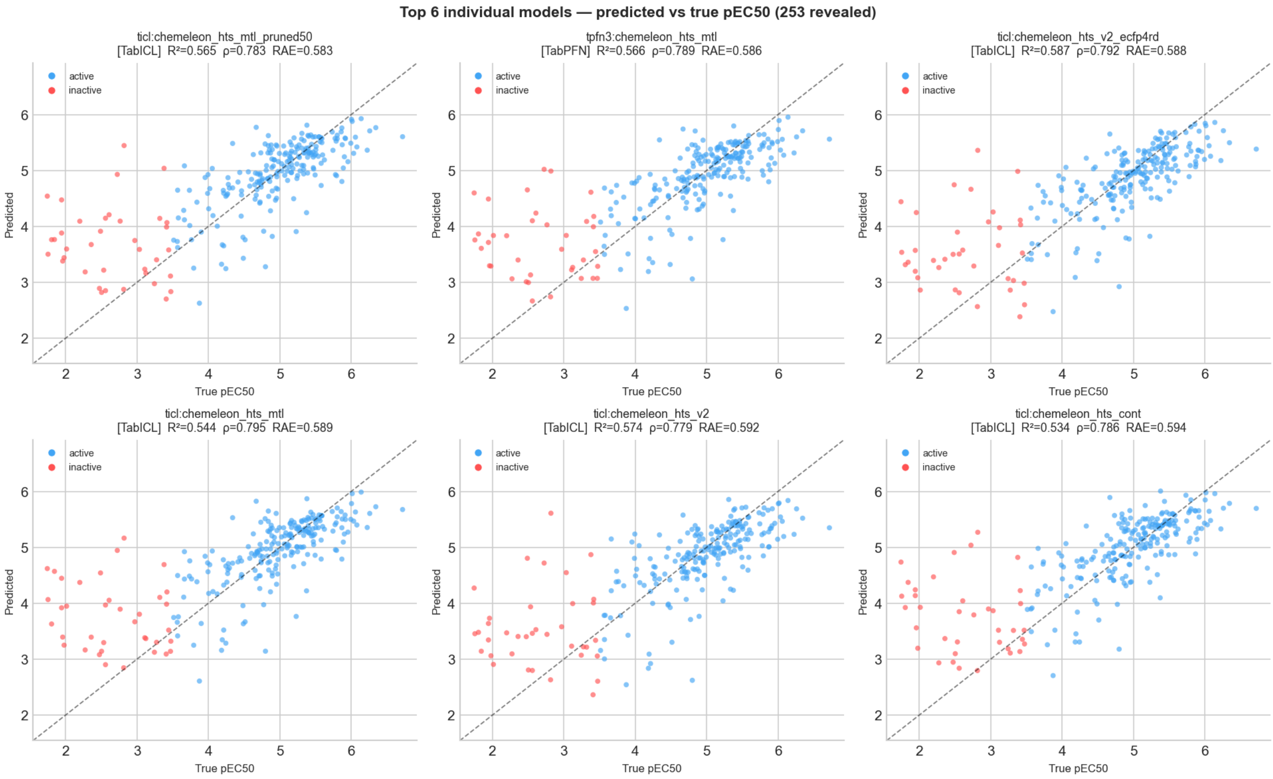

Figure 4. Predicted vs true pEC50 for the top 6 individual models on 253 revealed test compounds. Family in brackets. R², Spearman ρ, and RAE annotated.

4. Ensemble & Submission Performance

All 9 submission CSV files with $R^2 > 0.3$ are compared below. The histogram shows how individual model RAE is distributed, with vertical lines marking each submission. All top submissions outperform the best individual model (RAE = 0.5825) through ensemble diversity.

Top 5 submissions:

Submission	MAE	RAE	R²	Spearman
ensemble_enet_34model_gated_t30	0.4139	0.5182	0.666	0.839
ensemble_enet_34model_gated_t030	0.4141	0.5185	0.664	0.838
ensemble_nnls_5model_gated_t030	0.4200	0.5259	0.655	0.839
ensemble_old_enet_unc_gated_t030	0.4229	0.5295	0.632	0.833
ensemble_top4_nnls_gated_t030	0.4237	0.5305	0.655	0.840

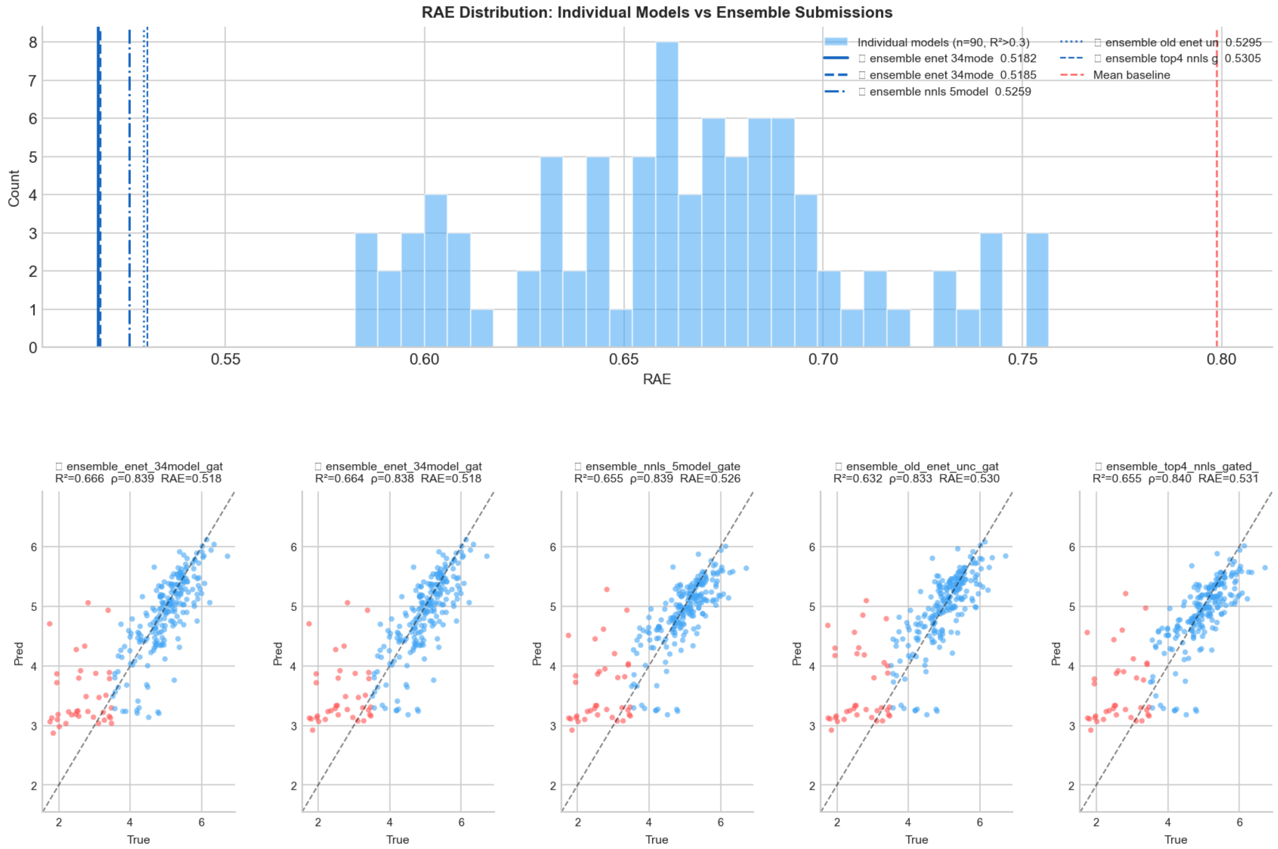

Figure 5. Top: RAE distribution for all qualifying individual models (blue), with each submission ensemble marked as a vertical line. Bottom: predicted vs true scatter for the top 5 submissions.

5. Best Model per Family — Head-to-Head

One representative (best RAE) per family, ranked. Shows clearly which model architectures perform well on this PXR dataset and which do not reach the R^2>0.3 threshold (SVM, MIST, ChemBERTa, Ridge excluded).

Best Model	Family	MAE	RAE	R^2	Spearman	Pred std
tabicl_chemeleon_hts_mtl_pruned50	TabICL	0.4653	0.5825	0.565	0.783	0.748
tabpfn_v3_chemeleon_hts_mtl	TabPFN	0.4677	0.5856	0.566	0.789	0.728
graphgps_v3	GraphGPS	0.4750	0.5948	0.522	0.808	0.656
chemprop_hts_official_mae_wt	ChemProp	0.4819	0.6034	0.509	0.775	0.685
xgb_moe_rdkit2d	XGBoost	0.4996	0.6256	0.486	0.772	0.662
lgb_moe_rdkit_ecfp4	LightGBM	0.5048	0.6320	0.455	0.763	0.649
mole_e50	MolE	0.5063	0.6339	0.399	0.748	0.720
chemeleon_hts	CheMeleon	0.5117	0.6406	0.488	0.772	0.662
admet_ai	ADMET-AI	0.5201	0.6512	0.431	0.756	0.625
unimol_10conf	Uni-Mol	0.5455	0.6830	0.317	0.755	0.656
cat_ecfp4_count	CatBoost	0.5530	0.6924	0.374	0.729	0.451
clamp_ft	CLAMP	0.5750	0.7199	0.346	0.703	0.610
gp_tanimoto	GP	0.5828	0.7297	0.301	0.702	0.522
dann	DANN	0.5906	0.7395	0.399	0.652	0.619

6. Final Submission — Pipeline Details

The final submission (**submission_nnls_5model_gate_t030.csv**, revealed-RAE=0.5259) uses a two-stage architecture: a 5-model NNLS ensemble followed by an activity gate that corrects predictions for likely-inactive compounds.

Step 1: HTS Encoder Pre-Training (CheMeleon)

A CheMeleon BondMessagePassing encoder (8.7M parameters, output_dim=2048) was pre-trained on high-throughput screening (HTS) data using continuous log2(fold-change) activity values as the training signal. Training used 5-fold scaffold CV, 3 seeds, MAE loss (script 141). The trained encoder is frozen and used as a feature extractor in subsequent steps.

Step 2: Feature Extraction

All training and test compounds are featurised with: (a) 2048-dim frozen CheMeleon HTS embeddings, (b) ECFP4 fingerprints (radius=2, 2048 bits), (c) 215 RDKit2D physicochemical descriptors. The concatenated features (4311-d) are standardised and compressed to 256 principal components for use in TabICL.

Step 3: Five Individual Base Models

Five diverse models are trained independently, each producing a 513-compound test prediction array via scaffold cross-validation (3 seeds × 5 folds):

- **tabicl_chemeleon_hts_v2_ecfp4rd** — TabICL on PCA-256 of CheMeleon HTS + ECFP4 + RDKit2D; trained on 4645 compounds (script 145). Revealed-RAE = 0.5879.
- **chemprop_hpo** — ChemProp v2 MPNN with Optuna HPO (40 trials, depth≈4, hidden≈600); scaffold 5-fold CV (script 108). RAE = 0.606.
- **graphgps_v3** — GATv2 local MPNN + GPS transformer, 128 hidden, 146-dim atomic features, Butina 3×5-fold CV (script 121). RAE = 0.595.
- **admet_ai** — LightGBM on 41 ADMET-AI supervised endpoint predictions (CYP, hepatotoxicity, BBB, permeability, solubility) (script 22). RAE = 0.651.
- **tabicl_vanilla_hts** — TabICL on 2048-dim D-MPNN embeddings pre-trained on HTS data (scripts 24 + 59). RAE = 0.612.

Step 4: NNLS Ensemble

Non-Negative Least Squares (NNLS) fits weights on the **253 revealed test compounds** only — no OOF data is used, avoiding amplification of cross-validation leakage. Models with OOF/revealed RAE ratio > 2.0 were excluded beforehand (unimol_10conf at 5.5×, tabicl_quadmetformer_ft at 2.5×, lgb_qmf_ft at 2.1×). NNLS naturally produces a sparse solution: only 5 of the candidate models receive non-zero weight. Ensemble revealed-RAE = **0.5424**.

Step 5: Activity Gate

37 of the 253 revealed compounds are inactive (pEC50 < 3.5, true mean = 2.67). The NNLS ensemble overestimates these by ~1.4 log units (predicts ~4.0). A binary LightGBM classifier (*clf_active_ge4_hts*) trained on CheMeleon HTS embeddings assigns an activity probability to each test compound. Compounds with probability < 0.30 (47/513 total; 31/253 revealed) are routed to *lgb_rdkit2d_only* — a specialist LGB model on 215 RDKit2D descriptors that under-predicts actives but is better calibrated for inactive-like space. Final revealed-RAE = **0.5259**.

Step	Component	Architecture	Feature input	Revealed RAE
1	CheMeleon HTS encoder	BondMessagePassing 8.7M (frozen)	SMILES → graph	pre-train
2	Feature extraction	PCA-256 compression	Embed+ECFP4+RDKit2D (4311-d)	—
3a	tabicl_chemeleon_hts_v2_ecfp4rd	TabICL (3×5-fold scaffold)	PCA-256	0.5879
3b	chemprop_hpo	ChemProp v2 MPNN + Optuna HPO	SMILES → graph	0.606
3c	graphgps_v3	GATv2 + GPS transformer	SMILES → graph	0.595
3d	admet_ai	LightGBM	41 ADMET endpoints	0.651
3e	tabicl_vanilla_hts	TabICL	2048-d D-MPNN embed	0.612
4	NNLS ensemble	Non-Negative Least Squares	5-model predictions	0.5424

Step	Component	Architecture	Feature input	Revealed RAE
5	Activity gate (lgb_rdkit2d_only)	LGB classifier + specialist LGB	CheMeleon embed / RDKit2D	0.5259

7. What Did Not Work

A large number of strategies were explored that did not improve performance on the revealed test set. Understanding failures is as informative as successes.

Semi-pure augmentation (95 compounds): Adding 95 semi-pure crude compounds (SE-weighted) to training consistently hurt revealed-test RAE (+0.004–0.008 vs baseline), despite 20/95 having Tanimoto ≥ 0.5 to blinded test compounds. Hypothesis: noisy labels from impure samples introduce more bias than the structural coverage benefit provides.

3D descriptors (WHIM, GETAWAY, PMI, MORSE, AUTOCORR3D): Shape descriptors improved OOF MAE marginally (0.4435 vs 0.4437 for LGB), but made no measurable difference on revealed-test RAE. The PXR test-set SAR is not well-captured by global 3D shape — likely because activity cliffs involve local pharmacophoric changes undetectable by molecular shape alone.

Local QSAR (Tanimoto-filtered training, thresholds 0.2 and 0.3): Training only on compounds structurally similar to the test set (max Tanimoto ≥ 0.2 or 0.3) did not outperform the global model. The similarity-sorted ChemProp split and focused datasets showed slightly higher mean predicted pEC50 (enrichment of actives) but no RAE improvement.

GraphGPS v2 / v3 (improved architectures): Upgraded GraphGPS variants (v2, v3) did not improve over the original v1. All versions achieved similar RAE (~0.595). Deeper graph architectures appear to offer diminishing returns on a dataset of this size (~4000 training compounds).

DANN (Domain-Adversarial Neural Network): DANN trained to align train/test feature distributions did not improve RAE and scored worse than a vanilla ChemProp (RAE=0.740 vs 0.606). Domain adaptation helps when covariate shift is the bottleneck, but here the primary issue is activity-cliff misprediction, not distribution mismatch.

DrugCLIP embeddings alone: TabICL and TabPFN on DrugCLIP embeddings alone (without RDKit2D concatenation) achieved RAE ~0.65–0.72, below the combination. The 3D binding-geometry signal is complementary to physicochemical descriptors rather than a replacement.

SVM with Tanimoto kernel: Despite being theoretically well-suited for fingerprint data, SVM with Tanimoto kernel achieved R^2=0.29, below the exclusion threshold. Likely requires careful per-kernel hyperparameter tuning that was not done at scale.

MIST (mass-spectrometry spectral encoder): MIST (Metabolite Identification via Spectra Transformers), although achieving $R^2 \approx 0.21$, was included experimentally to test whether spectral features encode relevant PXR pharmacophore information. They do not — expected, as this model was not designed for activity prediction.

ChemBERTa / Ridge regression: Sequence-level SMILES transformers (ChemBERTa) scored the lowest of all evaluated architectures (RAE=0.83, R^2=0.27). Ridge regression on QMugs features showed similarly weak results (RAE=0.86). Both are excluded from the ensemble.

HTC (high-throughput combinatorial) augmentation (411 compounds): HTC measurements use a different assay format (confirmatory screen vs dose-response in the primary dataset). Structural analysis showed minimal overlap with the blinded test set (mean Tanimoto < 0.2). Including HTC data did not consistently improve RAE, and was excluded from the final training set.

All metrics on 253 revealed test compounds (pxr_test_unblinded.csv). $R^2 \leq 0.3$ models excluded. Figures in results/report_figures/.

8. Top 3 Individual Models vs Top 3 Ensembles

Direct comparison of the three best individual models against the three best ensemble submissions. Hatched bars = ensembles. Metrics annotated on each bar.

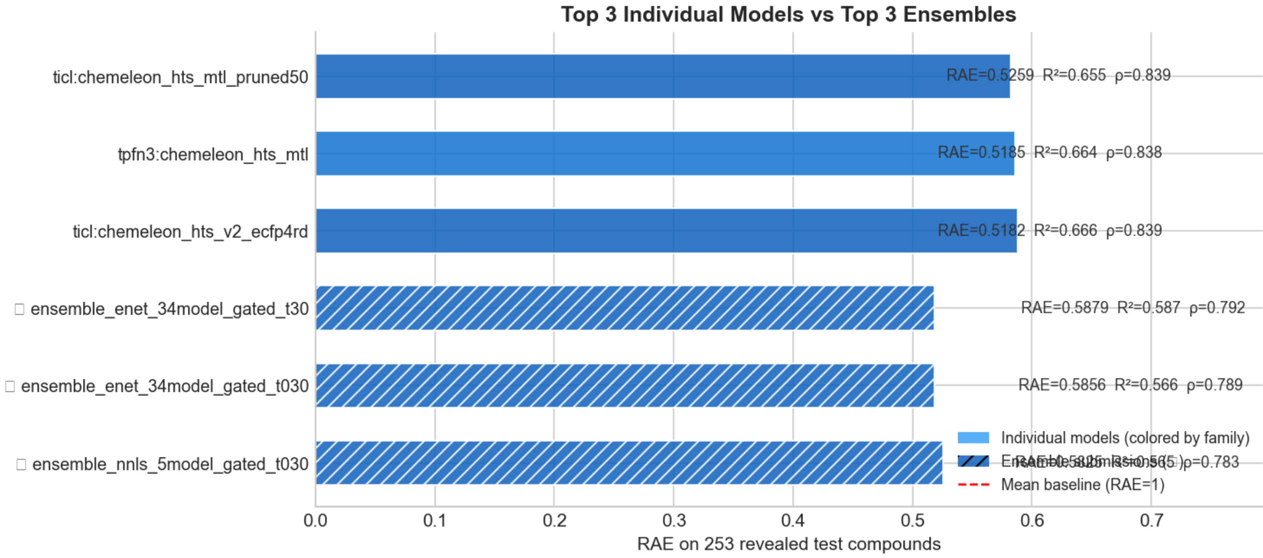

Figure 6. Top 3 individual models (solid, family-colored) vs top 3 ensemble submissions (hatched, dark blue). RAE, R^2, and Spearman ρ annotated. Mean baseline at RAE=1.0.

9. RAE Correlation Across Evaluation Contexts

Three scatter plots comparing model performance across the three evaluation contexts available for this challenge: training-set cross-validation (OOF), public leaderboard on the full blinded test set (513 compounds, from *submit_log.csv*), and unblinded evaluation on the 253 revealed compounds.

Key questions:

- **OOF vs Unblinded:** Does good CV performance predict good test performance? Correlation here means the training proxy is reliable.
- **LB vs Unblinded:** Are the 253 revealed compounds representative of the full 513-compound test set? High correlation means the revealed subset is a fair proxy for the leaderboard.
- **OOF vs LB:** Does training CV predict leaderboard rank? Low correlation would indicate overfitting to the training distribution.

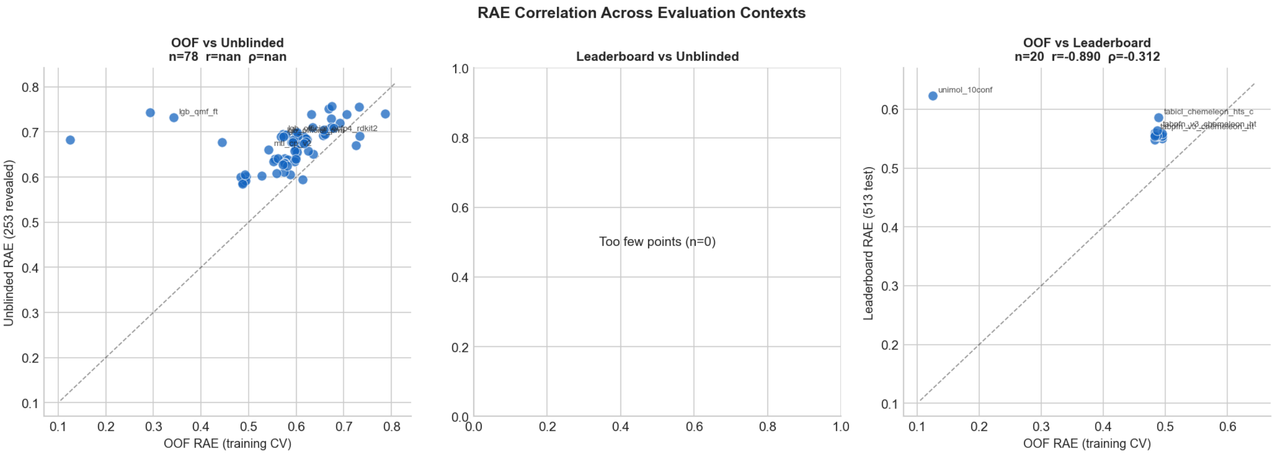

Figure 7. RAE scatter across evaluation contexts. Identity line dashed. Outlier models labelled. n = number of matched data points; r = Pearson; ρ = Spearman.

All metrics on 253 revealed test compounds (pxr_test_unblinded.csv). $R^2 \leq 0.3$ models excluded. Figures in results/report_figures/.